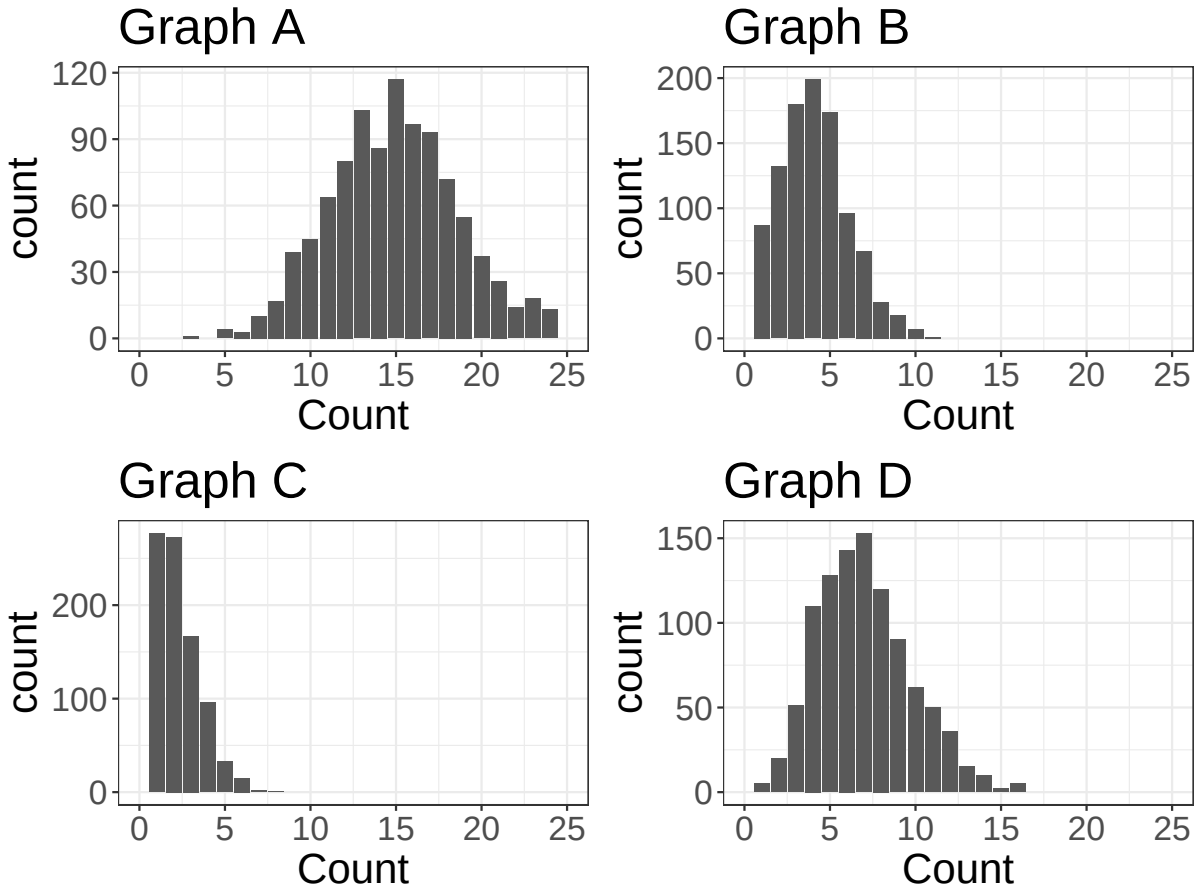
Activity

Group members:

Practice with the Poisson distribution

1. In the graph below, we have four Poisson distributions. Order the graphs from smallest λ to largest λ, and write down a guess at what you think λ might be for each graph.



2. Suppose we are counting the number of animals adopted from the Forsyth County animal shelter. We are told the number of dogs adopted per day follows a Poisson Distribution with $\lambda = 1.5$. What is the probability that on a randomly chosen day, *at most* 2 dogs are adopted?

Recall that if $Y \sim Poisson(\lambda)$, then $P(Y = y) = \dfrac{\lambda^y e^{-\lambda}}{y!}$

3. The Poisson distribution is for count data. Why is it ok for λ to not be a whole number?